



Syllabus: Sets, Relations, Functions, Partial orders, Lattices, **Monoids**, Groups.

Mark Distribution in Previous GATE

Year	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum	Average	Maximum
1 Mark Count	1	0	1	1	0	1	0	1	0	0.63	1
2 Marks Count	1	1	1	1	2	0	2	1	0	1.13	2
Total Marks	3	2	3	3	4	1	4	3	1	2.88	4

4.1

Binary Operation (8)

4.1.1 Binary Operation: GATE CSE 1989 | Question: 1-v



The number of possible commutative binary operations that can be defined on a set of n elements (for a given n) is _____.

gate1989 descriptive set-theory&algebra binary-operation

Answer key

4.1.2 Binary Operation: GATE CSE 1994 | Question: 2.2



On the set N of non-negative integers, the binary operation _____ is associative and non-commutative.

gate1994 set-theory&algebra normal group-theory binary-operation fill-in-the-blanks

Answer key

4.1.3 Binary Operation: GATE CSE 2003 | Question: 38



Consider the set $\{a, b, c\}$ with binary operators $+$ and $*$ defined as follows:

$+$	a	b	c
a	b	a	c
b	a	b	c
c	a	c	b

$*$	a	b	c
a	a	b	c
b	b	c	a
c	c	c	b

For example, $a + c = c$, $c + a = a$, $c * b = c$ and $b * c = a$.

Given the following set of equations:

- $(a * x) + (a * y) = c$
- $(b * x) + (c * y) = c$

The number of solution(s) (i.e., pair(s) (x, y) that satisfy the equations) is

- A. 0 B. 1 C. 2 D. 3

gatecse-2003 set-theory&algebra normal binary-operation

Answer key

4.1.4 Binary Operation: GATE CSE 2006 | Question: 28



A logical binary relation $\odot$, is defined as follows:

A	B	$A \odot B$
True	True	True
True	False	True
False	True	False
False	False	True

Let $\sim$ be the unary negation (NOT) operator, with higher precedence than $\odot$.

Which one of the following is equivalent to $A \wedge B$?

- A. $(\sim A \odot B)$
 C. $\sim(\sim A \odot \sim B)$

- B. $\sim(A \odot \sim B)$
 D. $\sim(\sim A \odot B)$

gatecse-2006 set-theory&algebra binary-operation propositional-logic mathematical-logic

Answer key

4.1.5 Binary Operation: GATE CSE 2013 | Question: 1



A binary operation $\oplus$ on a set of integers is defined as $x \oplus y = x^2 + y^2$. Which one of the following statements is **TRUE** about $\oplus$?

- A. Commutative but not associative
 C. Associative but not commutative
 B. Both commutative and associative
 D. Neither commutative nor associative

gatecse-2013 set-theory&algebra easy binary-operation

Answer key

4.1.6 Binary Operation: GATE CSE 2015 Set 1 | Question: 28



The binary operator $\neq$ is defined by the following truth table.

p	q	$p \neq q$
0	0	0
0	1	1
1	0	1
1	1	0

Which one of the following is true about the binary operator $\neq$?

- A. Both commutative and associative
 C. Not commutative but associative
 B. Commutative but not associative
 D. Neither commutative nor associative

gatecse-2015-set1 set-theory&algebra easy binary-operation boolean-algebra

Answer key

4.1.7 Binary Operation: GATE CSE 2015 Set 3 | Question: 2



Let $\#$ be the binary operator defined as

$X \# Y = X' + Y'$ where X and Y are Boolean variables.

Consider the following two statements.

- $(S_1) (P \# Q) \# R = P \# (Q \# R)$
- $(S_2) Q \# R = (R \# Q)$

Which are the following is/are true for the Boolean variables P, Q and R ?

- A. Only S_1 is true
 C. Both S_1 and S_2 are true
 B. Only S_2 is true
 D. Neither S_1 nor S_2 are true

gatecse-2015-set3 set-theory&algebra binary-operation normal

Answer key

4.1.8 Binary Operation: GATE IT 2006 | Question: 2



For the set N of natural numbers and a binary operation $f : N \times N \rightarrow N$, an element $z \in N$ is called an identity for f , if $f(a, z) = a = f(z, a)$, for all $a \in N$. Which of the following binary operations have an identity?

- $f(x, y) = x + y - 3$
- $f(x, y) = \max(x, y)$
- $f(x, y) = x^y$

- A. I and II only
 B. II and III only
 C. I and III only
 D. None of these

gateit-2006 set-theory&algebra easy binary-operation

Answer key

4.2.1 Countable Uncountable Set: GATE CSE 1994 | Question: 3.9



Every subset of a countable set is countable.

State whether the above statement is true or false with reason.

gate1994 set-theory&algebra normal set-theory countable-uncountable-set true-false

Answer key

4.2.2 Countable Uncountable Set: GATE CSE 2018 | Question: 27



Let N be the set of natural numbers. Consider the following sets,

- P : Set of Rational numbers (positive and negative)
- Q : Set of functions from $\{0, 1\}$ to N
- R : Set of functions from N to $\{0, 1\}$
- S : Set of finite subsets of N

Which of the above sets are countable?

- A. Q and S only B. P and S only C. P and R only D. P, Q and S only

gatecse-2018 set-theory&algebra countable-uncountable-set normal two-marks

Answer key

4.3 Functions (29)

4.3.1 Functions: GATE CSE 1987 | Question: 9b



How many one-to-one functions are there from a set A with n elements onto itself?

gate1987 set-theory&algebra functions descriptive

Answer key

4.3.2 Functions: GATE CSE 1988 | Question: 13ii



If the set S has a finite number of elements, prove that if f maps S onto S , then f is one-to-one.

gate1988 descriptive set-theory&algebra functions

Answer key

4.3.3 Functions: GATE CSE 1989 | Question: 13c



Find the number of single valued functions from set A to another set B , given that the cardinalities of the sets A and B are m and n respectively.

gate1989 descriptive functions set-theory&algebra

Answer key

4.3.4 Functions: GATE CSE 1993 | Question: 8.6



Let A and B be sets with cardinalities m and n respectively. The number of one-one mappings from A to B , when $m < n$, is

- A. m^n B. nP_m C. mC_n D. nC_m
E. mP_n

gate1993 set-theory&algebra functions easy

Answer key

4.3.5 Functions: GATE CSE 1996 | Question: 1.3



Suppose X and Y are sets and $|X|$ and $|Y|$ are their respective cardinality. It is given that there are exactly 97 functions from X to Y . From this one can conclude that

- A. $|X| = 1, |Y| = 97$ B. $|X| = 97, |Y| = 1$
C. $|X| = 97, |Y| = 97$ D. None of the above

Answer key

4.3.6 Functions: GATE CSE 1996 | Question: 2.1



Let R denote the set of real numbers. Let $f: R \times R \rightarrow R \times R$ be a bijective function defined by $f(x, y) = (x + y, x - y)$. The inverse function of f is given by

A. $f^{-1}(x, y) = \left(\frac{1}{x+y}, \frac{1}{x-y}\right)$

B. $f^{-1}(x, y) = (x - y, x + y)$

C. $f^{-1}(x, y) = \left(\frac{x+y}{2}, \frac{x-y}{2}\right)$

D. $f^{-1}(x, y) = [2(x - y), 2(x + y)]$

Answer key

4.3.7 Functions: GATE CSE 1997 | Question: 13



Let F be the set of one-to-one functions from the set $\{1, 2, \dots, n\}$ to the set $\{1, 2, \dots, m\}$ where $m \geq n \geq 1$.

- How many functions are members of F ?
- How many functions f in F satisfy the property $f(i) = 1$ for some $i, 1 \leq i \leq n$?
- How many functions f in F satisfy the property $f(i) < f(j)$ for all $i, j, 1 \leq i < j \leq n$?

Answer key

4.3.8 Functions: GATE CSE 1998 | Question: 1.8



The number of functions from an m element set to an n element set is

A. $m + n$

B. m^n

C. n^m

D. $m * n$

Answer key

4.3.9 Functions: GATE CSE 2001 | Question: 2.3



Let $f: A \rightarrow B$ a function, and let E and F be subsets of A . Consider the following statements about images.

- $S_1: f(E \cup F) = f(E) \cup f(F)$
- $S_2: f(E \cap F) = f(E) \cap f(F)$

Which of the following is true about S_1 and S_2 ?

A. Only S_1 is correctB. Only S_2 is correctC. Both S_1 and S_2 are correctD. None of S_1 and S_2 is correct

Answer key

4.3.10 Functions: GATE CSE 2001 | Question: 4



Consider the function $h: N \times N \rightarrow N$ so that $h(a, b) = (2a + 1)2^b - 1$, where $N = \{0, 1, 2, 3, \dots\}$ is the set of natural numbers.

- Prove that the function h is an injection (one-one).
- Prove that it is also a Surjection (onto)

Answer key

4.3.11 Functions: GATE CSE 2003 | Question: 37



Let $f : A \rightarrow B$ be an injective (one-to-one) function. Define $g : 2^A \rightarrow 2^B$ as:

$g(C) = \{f(x) \mid x \in C\}$, for all subsets C of A .

Define $h : 2^B \rightarrow 2^A$ as: $h(D) = \{x \mid x \in A, f(x) \in D\}$, for all subsets D of B . Which of the following statements is always true?

- A. $g(h(D)) \subseteq D$
 C. $g(h(D)) \cap D = \phi$
 B. $g(h(D)) \supseteq D$
 D. $g(h(D)) \cap (B - D) \neq \phi$

gatecse-2003 set-theory&algebra functions difficult

Answer key

4.3.12 Functions: GATE CSE 2003 | Question: 39



Let $\Sigma = \{a, b, c, d, e\}$ be an alphabet. We define an encoding scheme as follows:

$g(a) = 3, g(b) = 5, g(c) = 7, g(d) = 9, g(e) = 11$.

Let p_i denote the i -th prime number ($p_1 = 2$).

For a non-empty string $s = a_1 \dots a_n$, where each $a_i \in \Sigma$, define $f(s) = \prod_{i=1}^n P_i^{g(a_i)}$.

For a non-empty sequence $\langle s_1, \dots, s_n \rangle$ of strings from Σ^+ , define $h(\langle s_1 \dots s_n \rangle) = \prod_{i=1}^n P_i^{f(s_i)}$

Which of the following numbers is the encoding, h , of a non-empty sequence of strings?

- A. $2^7 3^7 5^7$
 B. $2^8 3^8 5^8$
 C. $2^9 3^9 5^9$
 D. $2^{10} 3^{10} 5^{10}$

gatecse-2003 set-theory&algebra functions normal

Answer key

4.3.13 Functions: GATE CSE 2005 | Question: 43



Let $f : B \rightarrow C$ and $g : A \rightarrow B$ be two functions and let $h = f \circ g$. Given that h is an onto function which one of the following is TRUE?

- A. f and g should both be onto functions
 C. g should be onto but f need not be onto
 B. f should be onto but g need not to be onto
 D. both f and g need not be onto

gatecse-2005 set-theory&algebra functions normal

Answer key

4.3.14 Functions: GATE CSE 2006 | Question: 2



Let X, Y, Z be sets of sizes x, y and z respectively. Let $W = X \times Y$ and E be the set of all subsets of W . The number of functions from Z to E is

- A. z^{2^y}
 B. $z \times 2^{xy}$
 C. $z^{2^{xy}}$
 D. 2^{xyz}

gatecse-2006 set-theory&algebra normal functions

Answer key

4.3.15 Functions: GATE CSE 2006 | Question: 25



Let $S = \{1, 2, 3, \dots, m\}, m > 3$. Let $X_1, \dots, X_n$ be subsets of S each of size 3. Define a function f from S to the set of natural numbers as, $f(i)$ is the number of sets X_j that contain the element i . That is $f(i) = |\{j \mid i \in X_j\}|$ then $\sum_{i=1}^m f(i)$ is:

- A. $3m$
 B. $3n$
 C. $2m + 1$
 D. $2n + 1$

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Answer key

4.3.16 Functions: GATE CSE 2007 | Question: 3



What is the maximum number of different Boolean functions involving n Boolean variables?

- A. n^2
 B. 2^n
 C. 2^{2^n}
 D. 2^{n^2}

gatecse-2007 combinatory functions normal

Answer key

4.3.17 Functions: GATE CSE 2012 | Question: 37



How many onto (or surjective) functions are there from an n -element ($n \geq 2$) set to a 2-element set?

- A. 2^n B. $2^n - 1$ C. $2^n - 2$ D. $2(2^n - 2)$

gatecse-2012 set-theory&algebra functions normal

Answer key

4.3.18 Functions: GATE CSE 2014 Set 1 | Question: 50



Let S denote the set of all functions $f : \{0, 1\}^4 \rightarrow \{0, 1\}$. Denote by N the number of functions from S to the set $\{0, 1\}$. The value of $\log_2 \log_2 N$ is _____.

gatecse-2014-set1 set-theory&algebra functions combinatory numerical-answers

Answer key

4.3.19 Functions: GATE CSE 2014 Set 3 | Question: 2



Let X and Y be finite sets and $f : X \rightarrow Y$ be a function. Which one of the following statements is TRUE?

- A. For any subsets A and B of X , $|f(A \cup B)| = |f(A)| + |f(B)|$
B. For any subsets A and B of X , $f(A \cap B) = f(A) \cap f(B)$
C. For any subsets A and B of X , $|f(A \cap B)| = \min\{|f(A)|, |f(B)|\}$
D. For any subsets S and T of Y , $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$

gatecse-2014-set3 set-theory&algebra functions normal

Answer key

4.3.20 Functions: GATE CSE 2014 Set 3 | Question: 49



Consider the set of all functions $f : \{0, 1, \dots, 2014\} \rightarrow \{0, 1, \dots, 2014\}$ such that $f(f(i)) = i$, for all $0 \leq i \leq 2014$. Consider the following statements:

P . For each such function it must be the case that for every i , $f(i) = i$.

Q . For each such function it must be the case that for some i , $f(i) = i$.

R . Each function must be onto.

Which one of the following is CORRECT?

- A. P , Q and R are true B. Only Q and R are true
C. Only P and Q are true D. Only R is true

gatecse-2014-set3 set-theory&algebra functions normal

Answer key

4.3.21 Functions: GATE CSE 2015 Set 1 | Question: 5



If $g(x) = 1 - x$ and $h(x) = \frac{x}{x-1}$, then $\frac{g(h(x))}{h(g(x))}$ is:

- A. $\frac{h(x)}{g(x)}$ B. $\frac{-1}{x}$ C. $\frac{g(x)}{h(x)}$ D. $\frac{x}{(1-x)^2}$

gatecse-2015-set1 set-theory&algebra functions normal

Answer key

4.3.22 Functions: GATE CSE 2015 Set 2 | Question: 40



The number of onto functions (surjective functions) from set $X = \{1, 2, 3, 4\}$ to set $Y = \{a, b, c\}$ is _____.

gatecse-2015-set2 set-theory&algebra functions normal numerical-answers

Answer key

4.3.23 Functions: GATE CSE 2015 Set 2 | Question: 54



Let X and Y denote the sets containing 2 and 20 distinct objects respectively and F denote the set of all possible functions defined from X to Y . Let f be randomly chosen from F . The probability of f being one-to-one is _____.

gatecse-2015-set2 set-theory&algebra functions normal numerical-answers

Answer key

4.3.24 Functions: GATE CSE 2016 Set 1 | Question: 28



A function $f: \mathbb{N}^+ \rightarrow \mathbb{N}^+$, defined on the set of positive integers $\mathbb{N}^+$, satisfies the following properties:

$$f(n) = f(n/2) \text{ if } n \text{ is even}$$

$$f(n) = f(n+5) \text{ if } n \text{ is odd}$$

Let $R = \{i \mid \exists j: f(j) = i\}$ be the set of distinct values that f takes. The maximum possible size of R is _____.

gatecse-2016-set1 set-theory&algebra functions normal numerical-answers

Answer key

4.3.25 Functions: GATE CSE 2021 Set 2 | Question: 11



Consider the following sets, where $n \geq 2$:

- S_1 : Set of all $n \times n$ matrices with entries from the set $\{a, b, c\}$
- S_2 : Set of all functions from the set $\{0, 1, 2, \dots, n^2 - 1\}$ to the set $\{0, 1, 2\}$

Which of the following choice(s) is/are correct?

- A. There does not exist a bijection from S_1 to S_2
- B. There exists a surjection from S_1 to S_2
- C. There exists a bijection from S_1 to S_2
- D. There does not exist an injection from S_1 to S_2

gatecse-2021-set2 multiple-selects set-theory&algebra functions one-mark

Answer key

4.3.26 Functions: GATE CSE 2023 | Question: 39



Let $f: A \rightarrow B$ be an onto (or surjective) function, where A and B are nonempty sets. Define an equivalence relation $\sim$ on the set A as

$$a_1 \sim a_2 \text{ if } f(a_1) = f(a_2),$$

where $a_1, a_2 \in A$. Let $\mathcal{E} = \{[x] : x \in A\}$ be the set of all the equivalence classes under $\sim$. Define a new mapping $F: \mathcal{E} \rightarrow B$ as

$$F([x]) = f(x), \quad \text{for all the equivalence classes } [x] \text{ in } \mathcal{E}.$$

Which of the following statements is/are TRUE?

- A. F is NOT well-defined.
- B. F is an onto (or surjective) function.
- C. F is a one-to-one (or injective) function.
- D. F is a bijective function.

gatecse-2023 set-theory&algebra functions multiple-selects two-marks

Answer key

4.3.27 Functions: GATE CSE 2024 | Set 1 | Question: 22



Let A and B be non-empty finite sets such that there exist one-to-one and onto functions (i) from A to B and (ii) from $A \times A$ to $A \cup B$. The number of possible values of $|A|$ is _____.

Answer key

4.3.28 Functions: GATE IT 2005 | Question: 31



Let f be a function from a set A to a set B , g a function from B to C , and h a function from A to C , such that $h(a) = g(f(a))$ for all $a \in A$. Which of the following statements is always true for all such functions f and g ?

- A. g is onto $\implies h$ is onto
 B. h is onto $\implies f$ is onto
 C. h is onto $\implies g$ is onto
 D. h is onto $\implies f$ and g are onto

gateit-2005 set-theory&algebra functions normal

Answer key

4.3.29 Functions: GATE IT 2006 | Question: 6



Given a boolean function $f(x_1, x_2, \dots, x_n)$, which of the following equations is NOT true?

- A. $f(x_1, x_2, \dots, x_n) = x'_1 f(x_1, x_2, \dots, x_n) + x_1 f(x_1, x_2, \dots, x_n)$
 B. $f(x_1, x_2, \dots, x_n) = x'_2 f(x_1, x_2, \dots, x_n) + x_2 f(x_1, x_2, \dots, x_n)$
 C. $f(x_1, x_2, \dots, x_n) = x'_n f(x_1, x_2, \dots, 0) + x_n f(x_1, x_2, \dots, 1)$
 D. $f(x_1, x_2, \dots, x_n) = f(0, x_2, \dots, x_n) + f(1, x_2, \dots, x_n)$

gateit-2006 set-theory&algebra functions normal

Answer key

4.4

Group Theory (33)

4.4.1 Group Theory: GATE CSE 1988 | Question: 2xviii



Show that if G is a group such that $(a * b)^2 = a^2 * b^2$ for all a, b belonging to G , then G is an abelian.

gate1988 descriptive group-theory

Answer key

4.4.2 Group Theory: GATE CSE 1990 | Question: 2-x



Match the pairs in the following questions:

(a) Groups	(p) Associativity
(b) Semigroups	(q) Identity
(c) Monoids	(r) Commutativity
(d) Abelian groups	(s) Left inverse

gate1990 match-the-following set-theory&algebra group-theory easy

Answer key

4.4.3 Group Theory: GATE CSE 1992 | Question: 14a



If G is a group of even order, then show that there exists an element $a \neq e$, the identity in G , such that $a^2 = e$.

gate1992 set-theory&algebra group-theory normal descriptive proof

Answer key

4.4.4 Group Theory: GATE CSE 1993 | Question: 28



Let $(\{p, q\}, *)$ be a semigroup where $p * p = q$. Show that:

- a. $p * q = q * p$ and
 b. $q * q = q$

gate1993 set-theory&algebra group-theory normal descriptive

Answer key

4.4.5 Group Theory: GATE CSE 1994 | Question: 1.10



Some group (G, o) is known to be abelian. Then, which one of the following is true for G ?

- A. $g = g^{-1}$ for every $g \in G$
- B. $g = g^2$ for every $g \in G$
- C. $(goh)^2 = g^2oh^2$ for every $g, h \in G$
- D. G is of finite order

gate1994 set-theory&algebra group-theory normal

Answer key

4.4.6 Group Theory: GATE CSE 1995 | Question: 2.17



Let A be the set of all non-singular matrices over real number and let $*$ be the matrix multiplication operation. Then

- A. A is closed under $*$ but $\langle A, * \rangle$ is not a semigroup.
- B. $\langle A, * \rangle$ is a semigroup but not a monoid.
- C. $\langle A, * \rangle$ is a monoid but not a group.
- D. $\langle A, * \rangle$ is a group but not an abelian group.

gate1995 set-theory&algebra group-theory

Answer key

4.4.7 Group Theory: GATE CSE 1995 | Question: 21



Let G_1 and G_2 be subgroups of a group G .

- A. Show that $G_1 \cap G_2$ is also a subgroup of G .
- B. Is $G_1 \cup G_2$ always a subgroup of G ?

gate1995 set-theory&algebra group-theory normal descriptive proof

Answer key

4.4.8 Group Theory: GATE CSE 1996 | Question: 1.4



Which of the following statements is FALSE?

- A. The set of rational numbers is an abelian group under addition
- B. The set of integers is an abelian group under addition
- C. The set of rational numbers form an abelian group under multiplication
- D. The set of real numbers excluding zero is an abelian group under multiplication

gate1996 set-theory&algebra group-theory normal

Answer key

4.4.9 Group Theory: GATE CSE 1996 | Question: 2.4



Which one of the following is false?

- A. The set of all bijective functions on a finite set forms a group under function composition
- B. The set $\{1, 2, \dots, p-1\}$ forms a group under multiplication mod p , where p is a prime number
- C. The set of all strings over a finite alphabet forms a group under concatenation
- D. A subset $S \neq \emptyset$ of G is a subgroup of the group $\langle G, * \rangle$ if and only if for any pair of elements $a, b \in S, a * b^{-1} \in S$

gate1996 set-theory&algebra normal set-theory group-theory

Answer key

4.4.10 Group Theory: GATE CSE 1997 | Question: 3.1



Let $(Z, *)$ be an algebraic structure where Z is the set of integers and the operation $*$ is defined by $n * m = \max(n, m)$. Which of the following statements is true for $(Z, *)$?

- A. $(Z, *)$ is a monoid
- B. $(Z, *)$ is an Abelian group

C. $(Z, *)$ is a group

D. None of the above

gate1997 set-theory&algebra group-theory normal

Answer key

4.4.11 Group Theory: GATE CSE 1998 | Question: 12



Let $(A, *)$ be a semigroup, Furthermore, for every a and b in A , if $a \neq b$, then $a * b \neq b * a$.

- Show that for every a in A , $a * a = a$
- Show that for every a, b in A , $a * b * a = a$
- Show that for every a, b, c in A , $a * b * c = a * c$

gate1998 set-theory&algebra group-theory descriptive

Answer key

4.4.12 Group Theory: GATE CSE 1999 | Question: 4



Let G be a finite group and H be a subgroup of G . For $a \in G$, define $aH = \{ah \mid h \in H\}$.

- Show that $|aH| = |bH|$.
- Show that for every pair of elements $a, b \in G$, either $aH = bH$ or aH and bH are disjoint.
- Use the above to argue that the order of H must divide the order of G .

gate1999 set-theory&algebra group-theory descriptive proof

Answer key

4.4.13 Group Theory: GATE CSE 2000 | Question: 4



Let $S = \{0, 1, 2, 3, 4, 5, 6, 7\}$ and $\otimes$ denote multiplication modulo 8, that is, $x \otimes y = (xy) \pmod 8$

- Prove that $(\{0, 1\}, \otimes)$ is not a group.
- Write three distinct groups $(G, \otimes)$ where $G \subset S$ and G has 2 elements.

gatecse-2000 set-theory&algebra descriptive group-theory

Answer key

4.4.14 Group Theory: GATE CSE 2002 | Question: 1.6



Which of the following is true?

- The set of all rational negative numbers forms a group under multiplication.
- The set of all non-singular matrices forms a group under multiplication.
- The set of all matrices forms a group under multiplication.
- Both B and C are true.

gatecse-2002 set-theory&algebra group-theory normal

Answer key

4.4.15 Group Theory: GATE CSE 2003 | Question: 7



Consider the set Σ^* of all strings over the alphabet $\Sigma = \{0, 1\}$. Σ^* with the concatenation operator for strings

- does not form a group
- forms a non-commutative group
- does not have a right identity element
- forms a group if the empty string is removed from Σ^*

gatecse-2003 set-theory&algebra group-theory normal

Answer key

4.4.16 Group Theory: GATE CSE 2004 | Question: 72



The following is the incomplete operation table of a 4–element group.

*	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b				
c				

The last row of the table is

- A. *caeb*
C. *cbea*
- B. *cbae*
D. *ceab*

gatecse-2004 set-theory&algebra group-theory normal

Answer key

4.4.17 Group Theory: GATE CSE 2005 | Question: 13



The set $\{1, 2, 4, 7, 8, 11, 13, 14\}$ is a group under multiplication modulo 15. The inverses of 4 and 7 are respectively:

- A. 3 and 13 B. 2 and 11 C. 4 and 13 D. 8 and 14

gatecse-2005 set-theory&algebra normal group-theory

Answer key

4.4.18 Group Theory: GATE CSE 2005 | Question: 46



Consider the set H of all 3×3 matrices of the type

$$\begin{pmatrix} a & f & e \\ 0 & b & d \\ 0 & 0 & c \end{pmatrix}$$

where a, b, c, d, e and f are real numbers and $abc \neq 0$. Under the matrix multiplication operation, the set H is:

- A. a group
B. a monoid but not a group
C. a semi group but not a monoid
D. neither a group nor a semi group

gatecse-2005 set-theory&algebra group-theory normal

Answer key 

4.4.19 Group Theory: GATE CSE 2006 | Question: 3



The set $\{1, 2, 3, 5, 7, 8, 9\}$ under multiplication modulo 10 is not a group. Given below are four possible reasons. Which one of them is false?

- A. It is not closed
B. 2 does not have an inverse
C. 3 does not have an inverse
D. 8 does not have an inverse

gatecse-2006 set-theory&algebra group-theory normal

Answer key 

4.4.20 Group Theory: GATE CSE 2007 | Question: 21



How many different non-isomorphic Abelian groups of order 4 are there?

- A. 2 B. 3 C. 4 D. 5

gatecse-2007 group-theory normal

Answer key 

4.4.21 Group Theory: GATE CSE 2009 | Question: 1



Which one of the following is **NOT** necessarily a property of a Group?

- A. Commutativity
C. Existence of inverse for every element

- B. Associativity
D. Existence of identity

gatecse-2009 set-theory&algebra easy group-theory

Answer key

4.4.22 Group Theory: GATE CSE 2009 | Question: 22



For the composition table of a cyclic group shown below:

*	a	b	c	d
a	a	b	c	d
b	b	a	d	c
c	c	d	b	a
d	d	c	a	b

Which one of the following choices is correct?

- A. a, b are generators
C. c, d are generators

- B. b, c are generators
D. d, a are generators

gatecse-2009 set-theory&algebra normal group-theory

Answer key

4.4.23 Group Theory: GATE CSE 2010 | Question: 4



Consider the set $S = \{1, \omega, \omega^2\}$, where ω and ω^2 are cube roots of unity. If $*$ denotes the multiplication operation, the structure $(S, *)$ forms

- A. A Group B. A Ring C. An integral domain D. A field

gatecse-2010 set-theory&algebra normal group-theory

Answer key

4.4.24 Group Theory: GATE CSE 2014 Set 3 | Question: 3



Let G be a group with 15 elements. Let L be a subgroup of G . It is known that $L \neq G$ and that the size of L is at least 4. The size of L is _____.

gatecse-2014-set3 set-theory&algebra group-theory numerical-answers normal

Answer key

4.4.25 Group Theory: GATE CSE 2014 Set 3 | Question: 50



There are two elements x, y in a group $(G, *)$ such that every element in the group can be written as a product of some number of x 's and y 's in some order. It is known that

$$x * x = y * y = x * y * x * y = y * x * y * x = e$$

where e is the identity element. The maximum number of elements in such a group is _____.

gatecse-2014-set3 set-theory&algebra group-theory numerical-answers normal

Answer key

4.4.26 Group Theory: GATE CSE 2018 | Question: 19



Let G be a finite group on 84 elements. The size of a largest possible proper subgroup of G is _____

gatecse-2018 group-theory numerical-answers set-theory&algebra one-mark

Answer key

4.4.27 Group Theory: GATE CSE 2019 | Question: 10



Let G be an arbitrary group. Consider the following relations on G :

- $R_1 : \forall a, b \in G, aR_1b$ if and only if $\exists g \in G$ such that $a = g^{-1}bg$
- $R_2 : \forall a, b \in G, aR_2b$ if and only if $a = b^{-1}$

Which of the above is/are equivalence relation/relations?

- A. R_1 and R_2 B. R_1 only C. R_2 only D. Neither R_1 nor R_2

gatecse-2019 engineering-mathematics discrete-mathematics set-theory&algebra group-theory one-mark

Answer key

4.4.28 Group Theory: GATE CSE 2020 | Question: 18



Let G be a group of 35 elements. Then the largest possible size of a subgroup of G other than G itself is _____.

gatecse-2020 numerical-answers group-theory easy one-mark

Answer key

4.4.29 Group Theory: GATE CSE 2021 Set 1 | Question: 34



Let G be a group of order 6, and H be a subgroup of G such that $1 < |H| < 6$. Which one of the following options is correct?

- A. Both G and H are always cyclic B. G may not be cyclic, but H is always cyclic
C. G is always cyclic, but H may not be cyclic D. Both G and H may not be cyclic

gatecse-2021-set1 set-theory&algebra group-theory two-marks

Answer key

4.4.30 Group Theory: GATE CSE 2022 | Question: 17



Which of the following statements is/are TRUE for a group G ?

- A. If for all $x, y \in G$, $(xy)^2 = x^2y^2$, then G is commutative.
B. If for all $x \in G$, $x^2 = 1$, then G is commutative. Here, 1 is the identity element of G .
C. If the order of G is 2, then G is commutative.
D. If G is commutative, then a subgroup of G need not be commutative.

gatecse-2022 set-theory&algebra group-theory multiple-selects one-mark

Answer key

4.4.31 Group Theory: GATE CSE 2023 | Question: 41



Let X be a set and 2^X denote the powerset of X .

Define a binary operation Δ on 2^X as follows:

$$A\Delta B = (A - B) \cup (B - A).$$

Let $H = (2^X, \Delta)$. Which of the following statements about H is/are correct?

- A. H is a group.
B. Every element in H has an inverse, but H is NOT a group.
C. For every $A \in 2^X$, the inverse of A is the complement of A .
D. For every $A \in 2^X$, the inverse of A is A .

gatecse-2023 set-theory&algebra group-theory multiple-selects two-marks

Answer key

4.4.32 Group Theory: GATE CSE 2024 | Set 1 | Question: 42



Consider the operators $\diamond$ and $\square$ defined by $a \diamond b = a + 2b$, $a \square b = ab$, for positive integers. Which of the following statements is/are TRUE?

- A. Operator $\diamond$ obeys the associative law

- B. Operator $\square$ obeys the associative law
 C. Operator $\diamond$ over the operator $\square$ obeys the distributive law
 D. Operator $\square$ over the operator $\diamond$ obeys the distributive law

gatecse2024-set1 multiple-selects set-theory&algebra group-theory two-marks

Answer key 

4.4.33 Group Theory: GATE CSE 2024 | Set 2 | Question: 53



Let Z_n be the group of integers $\{0, 1, 2, \dots, n-1\}$ with addition modulo n as the group operation. The number of elements in the group $Z_2 \times Z_3 \times Z_4$ that are their own inverses is _____.

gatecse2024-set2 numerical-answers set-theory&algebra group-theory two-marks

Answer key 

4.5 Identify Function (1)

4.5.1 Identify Function: GATE CSE 2025 | Set 1 | Question: 39



$A = \{0, 1, 2, 3, \dots\}$ is the set of non-negative integers. Let F be the set of functions from A to itself. For any two functions, $f_1, f_2 \in F$, we define

$$(f_1 \odot f_2)(n) = f_1(n) + f_2(n)$$

for every number n in A . Which of the following is/are CORRECT about the mathematical structure $(F, \odot)$?

- A. $(F, \odot)$ is an Abelian group.
 B. $(F, \odot)$ is an Abelian monoid.
 C. $(F, \odot)$ is a non-Abelian group.
 D. $(F, \odot)$ is a non-Abelian monoid.

gatecse2025-set1 set-theory&algebra identify-function group-theory multiple-selects easy two-marks

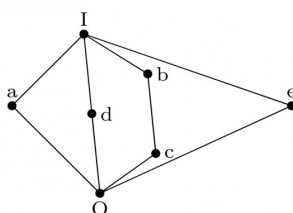
Answer key 

4.6 Lattice (10)

4.6.1 Lattice: GATE CSE 1988 | Question: 1vii



The complement(s) of the element ' a ' in the lattice shown in below figure is (are) _____



gate1988 descriptive lattice set-theory&algebra

Answer key 

4.6.2 Lattice: GATE CSE 1990 | Question: 17c



Show that the elements of the lattice $(N, \leq)$, where N is the set of positive integers and $a \leq b$ if and only if a divides b , satisfy the distributive property.

gate1990 descriptive set-theory&algebra lattice

Answer key 

4.6.3 Lattice: GATE CSE 1992 | Question: 14b



Consider the set of integers $\{1, 2, 3, 4, 6, 8, 12, 24\}$ together with the two binary operations LCM (lowest common multiple) and GCD (greatest common divisor). Which of the following algebraic structures does this represent?

A. group

B. ring

C. field

D. lattice

gate1992 set-theory&algebra partial-order lattice normal

Answer key

4.6.4 Lattice: GATE CSE 1994 | Question: 2.9



The Hasse diagrams of all the lattices with up to four elements are _____ (write all the relevant Hasse diagrams)

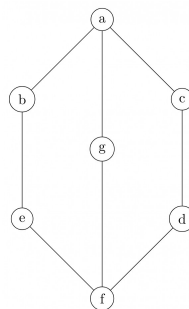
gate1994 set-theory&algebra lattice normal fill-in-the-blanks

Answer key

4.6.5 Lattice: GATE CSE 1997 | Question: 3.3



In the lattice defined by the Hasse diagram given in following figure, how many complements does the element 'e' have?



A. 2

B. 3

C. 0

D. 1

gate1997 set-theory&algebra lattice normal

Answer key

4.6.6 Lattice: GATE CSE 2002 | Question: 4



$S = \{(1,2), (2,1)\}$ is binary relation on set $A = \{1,2,3\}$. Is it irreflexive? Add the minimum number of ordered pairs to S to make it an equivalence relation. Give the modified S.

Let $S = \{a,b\}$ and let $\square(S)$ be the powerset of S. Consider the binary relation ' $\subseteq$ ' (set inclusion) on $\square(S)$. Draw the Hasse diagram corresponding to the lattice $(\square(S), \subseteq)$

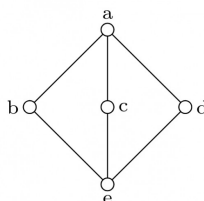
gatecse-2002 set-theory&algebra normal lattice descriptive

Answer key

4.6.7 Lattice: GATE CSE 2005 | Question: 9



The following is the Hasse diagram of the poset $\{a,b,c,d,e\}, \prec$



The poset is :

A. not a lattice

B. a lattice but not a distributive lattice

C. a distributive lattice but not a Boolean algebra

D. a Boolean algebra

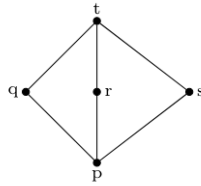
gatecse-2005 set-theory&algebra lattice normal

Answer key

4.6.8 Lattice: GATE CSE 2015 Set 1 | Question: 34



Suppose $L = \{p, q, r, s, t\}$ is a lattice represented by the following Hasse diagram:



For any $x, y \in L$, not necessarily distinct, $x \vee y$ and $x \wedge y$ are join and meet of x, y , respectively. Let $L^3 = \{(x, y, z) : x, y, z \in L\}$ be the set of all ordered triplets of the elements of L . Let p_r be the probability that an element $(x, y, z) \in L^3$ chosen equiprobably satisfies $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$. Then

- A. $p_r = 0$
- B. $p_r = 1$
- C. $0 < p_r \leq \frac{1}{5}$
- D. $\frac{1}{5} < p_r < 1$

gatecse-2015-set1 set-theory&algebra normal lattice

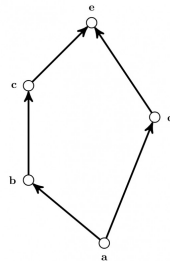
Answer key

4.6.9 Lattice: GATE CSE 2017 Set 2 | Question: 21



Consider the set $X = \{a, b, c, d, e\}$ under partial ordering $R = \{(a, a), (a, b), (a, c), (a, d), (a, e), (b, b), (b, c), (b, e), (c, c), (c, e), (d, d), (d, e), (e, e)\}$

The Hasse diagram of the partial order (X, R) is shown below.



The minimum number of ordered pairs that need to be added to R to make (X, R) a lattice is _____

gatecse-2017-set2 set-theory&algebra lattice numerical-answers normal

Answer key

4.6.10 Lattice: GATE IT 2008 | Question: 28



Consider the following Hasse diagrams.



Which all of the above represent a lattice?

- A. (i) and (iv) only
- B. (ii) and (iii) only
- C. (iii) only
- D. (i), (ii) and (iv) only

gateit-2008 set-theory&algebra lattice normal

Answer key

4.7.1 Mathematical Induction: GATE CSE 1995 | Question: 23



Prove using mathematical induction for $n \geq 5, 2^n > n^2$

gate1995 set-theory&algebra proof mathematical-induction descriptive

Answer key

4.7.2 Mathematical Induction: GATE CSE 2000 | Question: 3



Consider the following sequence:

$s_1 = s_2 = 1$ and $s_i = 1 + \min(s_{i-1}, s_{i-2})$ for $i > 2$.

Prove by induction on n that $s_n = \lceil \frac{n}{2} \rceil$.

gatecse-2000 set-theory&algebra mathematical-induction descriptive

Answer key

4.8 Number Theory (7)

4.8.1 Number Theory: GATE CSE 1995 | Question: 7(A)



Determine the number of divisors of 600.

gate1995 set-theory&algebra number-theory numerical-answers

Answer key

4.8.2 Number Theory: GATE CSE 2014 Set 2 | Question: 49



The number of distinct positive integral factors of 2014 is _____

gatecse-2014-set2 set-theory&algebra easy numerical-answers number-theory

Answer key

4.8.3 Number Theory: GATE CSE 2015 Set 2 | Question: 9



The number of divisors of 2100 is ____.

gatecse-2015-set2 set-theory&algebra number-theory easy numerical-answers

Answer key

4.8.4 Number Theory: GATE IT 2005 | Question: 34



Let $n = p^2q$, where p and q are distinct prime numbers. How many numbers m satisfy $1 \leq m \leq n$ and $\gcd(m, n) = 1$? Note that $\gcd(m, n)$ is the greatest common divisor of m and n .

- A. $p(q-1)$ B. pq
C. $(p^2-1)(q-1)$ D. $p(p-1)(q-1)$

gateit-2005 set-theory&algebra normal number-theory

Answer key

4.8.5 Number Theory: GATE IT 2007 | Question: 16



The minimum positive integer p such that $3^p \pmod{17} = 1$ is

- A. 5 B. 8 C. 12 D. 16

gateit-2007 set-theory&algebra normal number-theory

Answer key

4.8.6 Number Theory: GATE IT 2008 | Question: 24



The exponent of 11 in the prime factorization of $300!$ is

- A. 27 B. 28 C. 29 D. 30

gateit-2008 set-theory&algebra normal number-theory

Answer key

4.8.7 Number Theory: GATE1991-15,a



Show that the product of the least common multiple and the greatest common divisor of two positive integers a and b is $a \times b$.

gate1991 set-theory&algebra normal number-theory proof descriptive

Answer key

4.9

Onto (1)

4.9.1 Onto: GATE CSE 2025 | Set 1 | Question: 7



If $f(\cdot)$ and $f(g(\cdot))$ are onto (surjective) functions, which ONE of the following is TRUE about the function $g(\cdot)$?

- A. $g(\cdot)$ must be an onto (surjective) function.
- B. $g(\cdot)$ must be a one-to-one (injective) function.
- C. $g(\cdot)$ must be a bijective function, that is, both one-to-one and onto.
- D. $g(\cdot)$ is not required to be a one-to-one or onto function.

gatecse2025-set1 set-theory&algebra onto functions one-mark

Answer key

4.10

Partial Order (10)

4.10.1 Partial Order: GATE CSE 1991 | Question: 01,xiv



If the longest chain in a partial order is of length n , then the partial order can be written as a _____ of n antichains.

gate1991 set-theory&algebra partial-order normal fill-in-the-blanks

Answer key

4.10.2 Partial Order: GATE CSE 1993 | Question: 8.5



The less-than relation, $<$, on reals is

- A. a partial ordering since it is asymmetric and reflexive
- B. a partial ordering since it is antisymmetric and reflexive
- C. not a partial ordering because it is not asymmetric and not reflexive
- D. not a partial ordering because it is not antisymmetric and reflexive
- E. none of the above

gate1993 set-theory&algebra partial-order easy

Answer key

4.10.3 Partial Order: GATE CSE 1996 | Question: 1.2



Let $X = \{2, 3, 6, 12, 24\}$, Let $\leq$ be the partial order defined by $X \leq Y$ if x divides y . Number of edges in the Hasse diagram of $(X, \leq)$ is

- A. 3
- B. 4
- C. 9
- D. None of the above

gate1996 set-theory&algebra partial-order normal

Answer key

4.10.4 Partial Order: GATE CSE 1997 | Question: 6.1



A partial order $\leq$ is defined on the set $S = \{x, a_1, a_2, \dots, a_n, y\}$ as $x \leq_i a_i$ for all i and $a_i \leq y$ for all i , where $n \geq 1$. The number of total orders on the set S which contain the partial order $\leq$ is

- A. $n!$
- B. $n + 2$
- C. n
- D. 1

gate1997 set-theory&algebra partial-order normal

Answer key

4.10.5 Partial Order: GATE CSE 1998 | Question: 11



Suppose $A = \{a, b, c, d\}$ and Π_1 is the following partition of A

$$\Pi_1 = \{\{a, b, c\}, \{d\}\}$$

- List the ordered pairs of the equivalence relations induced by Π_1 .
- Draw the graph of the above equivalence relation.
- Let $\Pi_2 = \{\{a\}, \{b\}, \{c\}, \{d\}\}$

$$\Pi_3 = \{\{a, b, c, d\}\}$$

$$\text{and } \Pi_4 = \{\{a, b\}, \{c, d\}\}$$

Draw a Poset diagram of the poset, $\langle \{\Pi_1, \Pi_2, \Pi_3, \Pi_4\}, \text{refines} \rangle$.

gate1998 set-theory&algebra normal partial-order descriptive

Answer key

4.10.6 Partial Order: GATE CSE 2003 | Question: 31



Let $(S, \leq)$ be a partial order with two minimal elements a and b , and a maximum element c . Let $P: S \rightarrow \{\text{True}, \text{False}\}$ be a predicate defined on S . Suppose that $P(a) = \text{True}$, $P(b) = \text{False}$ and $P(x) \implies P(y)$ for all $x, y \in S$ satisfying $x \leq y$, where $\implies$ stands for logical implication. Which of the following statements CANNOT be true?

- $P(x) = \text{True}$ for all $x \in S$ such that $x \neq b$
- $P(x) = \text{False}$ for all $x \in S$ such that $x \neq a$ and $x \neq c$
- $P(x) = \text{False}$ for all $x \in S$ such that $b \leq x$ and $x \neq c$
- $P(x) = \text{False}$ for all $x \in S$ such that $a \leq x$ and $b \leq x$

gatecse-2003 set-theory&algebra partial-order normal propositional-logic

Answer key

4.10.7 Partial Order: GATE CSE 2004 | Question: 73



The inclusion of which of the following sets into

$$S = \{\{1, 2\}, \{1, 2, 3\}, \{1, 3, 5\}, \{1, 2, 4\}, \{1, 2, 3, 4, 5\}\}$$

is necessary and sufficient to make S a complete lattice under the partial order defined by set containment?

- $\{1\}$
- $\{1\}, \{2, 3\}$
- $\{1\}, \{1, 3\}$
- $\{1\}, \{1, 3\}, \{1, 2, 3, 4\}, \{1, 2, 3, 5\}$

gatecse-2004 set-theory&algebra partial-order normal

Answer key

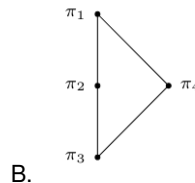
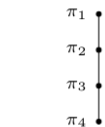
4.10.8 Partial Order: GATE CSE 2007 | Question: 26

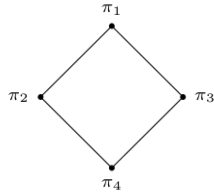


Consider the set $S = \{a, b, c, d\}$. Consider the following 4 partitions $\pi_1, \pi_2, \pi_3, \pi_4$ on

$$S: \pi_1 = \{\overline{abcd}\}, \pi_2 = \{\overline{ab}, \overline{cd}\}, \pi_3 = \{\overline{abc}, \overline{d}\}, \pi_4 = \{\overline{a}, \overline{b}, \overline{c}, \overline{d}\}.$$

Let $\prec$ be the partial order on the set of partitions $S' = \{\pi_1, \pi_2, \pi_3, \pi_4\}$ defined as follows: $\pi_i \prec \pi_j$ if and only if π_i refines π_j . The poset diagram for $(S', \prec)$ is:

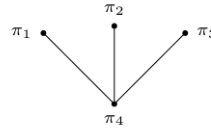




C.

gatecse-2007 set-theory&algebra normal partial-order descriptive

Answer key



D.

4.10.9 Partial Order: GATE CSE 2024 | Set 2 | Question: 24



Let P be the partial order defined on the set $\{1, 2, 3, 4\}$ as follows

$$P = \{(x, x) \mid x \in \{1, 2, 3, 4\}\} \cup \{(1, 2), (3, 2), (3, 4)\}$$

The number of total orders on $\{1, 2, 3, 4\}$ that contain P is _____.

gatecse2024-set2 numerical-answers set-theory&algebra partial-order one-mark

Answer key

4.10.10 Partial Order: GATE IT 2007 | Question: 23



A partial order P is defined on the set of natural numbers as follows. Here $\frac{x}{y}$ denotes integer division.

- $(0, 0) \in P$.
- $(a, b) \in P$ if and only if $(a \% 10) \leq (b \% 10)$ and $(\frac{a}{10}, \frac{b}{10}) \in P$.

Consider the following ordered pairs:

- $(101, 22)$
- $(22, 101)$
- $(145, 265)$
- $(0, 153)$

Which of these ordered pairs of natural numbers are contained in P ?

- A. (i) and (iii) B. (ii) and (iv) C. (i) and (iv) D. (iii) and (iv)

gateit-2007 set-theory&algebra partial-order normal

Answer key

4.11 Polynomials (4)

4.11.1 Polynomials: GATE CSE 1997 | Question: 4.4



A polynomial $p(x)$ is such that $p(0) = 5, p(1) = 4, p(2) = 9$ and $p(3) = 20$. The minimum degree it should have is

- A. 1 B. 2 C. 3 D. 4

gate1997 set-theory&algebra normal polynomials

Answer key

4.11.2 Polynomials: GATE CSE 2000 | Question: 2.4



A polynomial $p(x)$ satisfies the following:

- $p(1) = p(3) = p(5) = 1$
- $p(2) = p(4) = -1$

The minimum degree of such a polynomial is

A. 1

B. 2

C. 3

D. 4

gatecse-2000 set-theory&algebra normal polynomials

Answer key 

4.11.3 Polynomials: GATE CSE 2014 Set 2 | Question: 5



A non-zero polynomial $f(x)$ of degree 3 has roots at $x = 1$, $x = 2$ and $x = 3$. Which one of the following must be TRUE?

A. $f(0)f(4) < 0$

C. $f(0) + f(4) > 0$

B. $f(0)f(4) > 0$

D. $f(0) + f(4) < 0$

gatecse-2014-set2 set-theory&algebra polynomials normal

Answer key 

4.11.4 Polynomials: GATE CSE 2017 Set 2 | Question: 24



Consider the quadratic equation $x^2 - 13x + 36 = 0$ with coefficients in a base b . The solutions of this equation in the same base b are $x = 5$ and $x = 6$. Then $b = \underline{\hspace{2cm}}$

gatecse-2017-set2 polynomials numerical-answers set-theory&algebra

Answer key 

4.12

Relations (37)

4.12.1 Relations: GATE CSE 1987 | Question: 2d



State whether the following statements are TRUE or FALSE:

The union of two equivalence relations is also an equivalence relation.

gate1987 set-theory&algebra relations true-false

Answer key 

4.12.2 Relations: GATE CSE 1987 | Question: 9a



How many binary relations are there on a set A with n elements?

gate1987 set-theory&algebra relations descriptive

Answer key 

4.12.3 Relations: GATE CSE 1987 | Question: 9e



How many true inclusion relations are there of the form $A \subseteq B$, where A and B are subsets of a set S with n elements?

gate1987 set-theory&algebra relations descriptive

Answer key 

4.12.4 Relations: GATE CSE 1989 | Question: 1-iv



The transitive closure of the relation $\{(1,2), (2,3), (3,4), (5,4)\}$ on the set $\{1,2,3,4,5\}$ is $\underline{\hspace{2cm}}$.

gate1989 set-theory&algebra relations descriptive

Answer key 

4.12.5 Relations: GATE CSE 1992 | Question: 15.b



Let S be the set of all integers and let $n > 1$ be a fixed integer. Define for $a, b \in S$, aRb iff $a - b$ is a multiple of n . Show that R is an equivalence relation and find its equivalence classes for $n = 5$.

gate1992 set-theory&algebra normal relations descriptive

Answer key 

4.12.6 Relations: GATE CSE 1994 | Question: 2.3



Amongst the properties {reflexivity, symmetry, anti-symmetry, transitivity} the relation $R = \{(x, y) \in N^2 | x \neq y\}$ satisfies _____

gate1994 set-theory&algebra normal relations fill-in-the-blanks

Answer key

4.12.7 Relations: GATE CSE 1995 | Question: 1.19



Let R be a symmetric and transitive relation on a set A . Then

- A. R is reflexive and hence an equivalence relation
- B. R is reflexive and hence a partial order
- C. R is reflexive and hence not an equivalence relation
- D. None of the above

gate1995 set-theory&algebra relations normal

Answer key

4.12.8 Relations: GATE CSE 1996 | Question: 2.2



Let R be a non-empty relation on a collection of sets defined by $A R_B$ if and only if $A \cap B = \phi$. Then, (pick the true statement)

- A. A is reflexive and transitive
- B. R is symmetric and not transitive
- C. R is an equivalence relation
- D. R is not reflexive and not symmetric

gate1996 set-theory&algebra relations normal

Answer key

4.12.9 Relations: GATE CSE 1996 | Question: 8



Let F be the collection of all functions $f : \{1, 2, 3\} \rightarrow \{1, 2, 3\}$. If f and $g \in F$, define an equivalence relation $\sim$ by $f \sim g$ if and only if $f(3) = g(3)$.

- a. Find the number of equivalence classes defined by $\sim$.
- b. Find the number of elements in each equivalence class.

gate1996 set-theory&algebra relations functions normal descriptive

Answer key

4.12.10 Relations: GATE CSE 1997 | Question: 14



Let R be a reflexive and transitive relation on a set A . Define a new relation E on A as

$$E = \{(a, b) \mid (a, b) \in R \text{ and } (b, a) \in R\}$$

Prove that E is an equivalence relation on A .

Define a relation $\leq$ on the equivalence classes of E as $E_1 \leq E_2$ if $\exists a, b$ such that $a \in E_1, b \in E_2$ and $(a, b) \in R$. Prove that $\leq$ is a partial order.

gate1997 set-theory&algebra relations normal proof descriptive

Answer key

4.12.11 Relations: GATE CSE 1997 | Question: 6.3



The number of equivalence relations of the set $\{1, 2, 3, 4\}$ is

- A. 15
- B. 16
- C. 24
- D. 4

gate1997 set-theory&algebra relations normal

Answer key

4.12.12 Relations: GATE CSE 1998 | Question: 1.6



Suppose A is a finite set with n elements. The number of elements in the largest equivalence relation of A is

A. n B. n^2

C. 1

D. $n + 1$

gate1998 set-theory&algebra relations easy

Answer key

4.12.13 Relations: GATE CSE 1998 | Question: 1.7



Let R_1 and R_2 be two equivalence relations on a set. Consider the following assertions:

- $R_1 \cup R_2$ is an equivalence relation
- $R_1 \cap R_2$ is an equivalence relation

Which of the following is correct?

- Both assertions are true
- Assertion (i) is true but assertion (ii) is not true
- Assertion (ii) is true but assertion (i) is not true
- Neither (i) nor (ii) is true

gate1998 set-theory&algebra relations normal

Answer key

4.12.14 Relations: GATE CSE 1998 | Question: 10a



Prove by induction that the expression for the number of diagonals in a polygon of n sides is $\frac{n(n-3)}{2}$

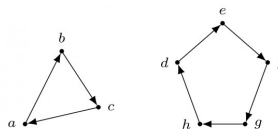
gate1998 set-theory&algebra descriptive relations

Answer key

4.12.15 Relations: GATE CSE 1998 | Question: 10b



Let R be a binary relation on $A = \{a, b, c, d, e, f, g, h\}$ represented by the following two component digraph. Find the smallest integers m and n such that $m < n$ and $R^m = R^n$.



gate1998 descriptive set-theory&algebra relations

Answer key

4.12.16 Relations: GATE CSE 1998 | Question: 2.3



The binary relation $R = \{(1,1), (2,1), (2,2), (2,3), (2,4), (3,1), (3,2), (3,3), (3,4)\}$ on the set $A = \{1, 2, 3, 4\}$ is

- reflexive, symmetric and transitive
- neither reflexive, nor irreflexive but transitive
- irreflexive, symmetric and transitive
- irreflexive and antisymmetric

gate1998 set-theory&algebra easy relations

Answer key

4.12.17 Relations: GATE CSE 1999 | Question: 1.2



The number of binary relations on a set with n elements is:

- n^2
- 2^n
- 2^{n^2}
- None of the above

gate1999 set-theory&algebra relations combinatory easy

Answer key

4.12.18 Relations: GATE CSE 1999 | Question: 2.3



Let L be a set with a relation R which is transitive, anti-symmetric and reflexive and for any two elements $a, b \in L$, let the least upper bound $\text{lub}(a, b)$ and the greatest lower bound $\text{glb}(a, b)$ exist. Which of the following

is/are true?

- A. L is a poset
- B. L is a Boolean algebra
- C. L is a lattice
- D. None of the above

gate1999 set-theory&algebra normal relations multiple-selects

Answer key

4.12.19 Relations: GATE CSE 1999 | Question: 3



- a. Mr. X claims the following:
If a relation R is both symmetric and transitive, then R is reflexive. For this, Mr. X offers the following proof:
"From xRy , using symmetry we get yRx . Now because R is transitive xRy and yRx together imply xRx . Therefore, R is reflexive".
- b. Give an example of a relation R which is symmetric and transitive but not reflexive.

gate1999 set-theory&algebra relations normal descriptive

Answer key

4.12.20 Relations: GATE CSE 2000 | Question: 2.5



A relation R is defined on the set of integers as xRy iff $(x + y)$ is even. Which of the following statements is true?

- A. R is not an equivalence relation
- B. R is an equivalence relation having 1 equivalence class
- C. R is an equivalence relation having 2 equivalence classes
- D. R is an equivalence relation having 3 equivalence classes

gatecse-2000 set-theory&algebra relations normal

Answer key

4.12.21 Relations: GATE CSE 2001 | Question: 1.2



Consider the following relations:

- $R_1(a, b)$ iff $(a + b)$ is even over the set of integers
- $R_2(a, b)$ iff $(a + b)$ is odd over the set of integers
- $R_3(a, b)$ iff $a.b > 0$ over the set of non-zero rational numbers
- $R_4(a, b)$ iff $|a - b| \leq 2$ over the set of natural numbers

Which of the following statements is correct?

- A. R_1 and R_2 are equivalence relations, R_3 and R_4 are not
- B. R_1 and R_3 are equivalence relations, R_2 and R_4 are not
- C. R_1 and R_4 are equivalence relations, R_2 and R_3 are not
- D. R_1, R_2, R_3 and R_4 all are equivalence relations

gatecse-2001 set-theory&algebra normal relations

Answer key

4.12.22 Relations: GATE CSE 2002 | Question: 2.17



The binary relation $S = \phi(\text{empty set})$ on a set $A = \{1, 2, 3\}$ is

- A. Neither reflexive nor symmetric
- B. Symmetric and reflexive
- C. Transitive and reflexive
- D. Transitive and symmetric

gatecse-2002 set-theory&algebra normal relations

Answer key

4.12.23 Relations: GATE CSE 2002 | Question: 3



Let A be a set of $n(> 0)$ elements. Let N_r be the number of binary relations on A and let N_f be the number of

functions from A to A

- A. Give the expression for N_r , in terms of n .
- B. Give the expression for N_f , terms of n .
- C. Which is larger for all possible n, N_r or N_f

gatecse-2002 set-theory&algebra normal descriptive relations

Answer key

4.12.24 Relations: GATE CSE 2004 | Question: 24



Consider the binary relation:

$$S = \{(x, y) \mid y = x + 1 \text{ and } x, y \in \{0, 1, 2\}\}$$

The reflexive transitive closure is S is

- A. $\{(x, y) \mid y > x \text{ and } x, y \in \{0, 1, 2\}\}$
- B. $\{(x, y) \mid y \geq x \text{ and } x, y \in \{0, 1, 2\}\}$
- C. $\{(x, y) \mid y < x \text{ and } x, y \in \{0, 1, 2\}\}$
- D. $\{(x, y) \mid y \leq x \text{ and } x, y \in \{0, 1, 2\}\}$

gatecse-2004 set-theory&algebra easy relations

Answer key

4.12.25 Relations: GATE CSE 2005 | Question: 42



Let R and S be any two equivalence relations on a non-empty set A . Which one of the following statements is TRUE?

- A. $R \cup S, R \cap S$ are both equivalence relations
- B. $R \cup S$ is an equivalence relation
- C. $R \cap S$ is an equivalence relation
- D. Neither $R \cup S$ nor $R \cap S$ are equivalence relations

gatecse-2005 set-theory&algebra normal relations

Answer key

4.12.26 Relations: GATE CSE 2005 | Question: 7



The time complexity of computing the transitive closure of a binary relation on a set of n elements is known to be:

- A. $O(n)$
- B. $O(n \log n)$
- C. $O\left(n^{\frac{3}{2}}\right)$
- D. $O(n^3)$

gatecse-2005 set-theory&algebra normal relations

Answer key

4.12.27 Relations: GATE CSE 2006 | Question: 4



A relation R is defined on ordered pairs of integers as follows:

$$(x, y)R(u, v) \text{ if } x < u \text{ and } y > v$$

Then R is:

- A. Neither a Partial Order nor an Equivalence Relation
- B. A Partial Order but not a Total Order
- C. A total Order
- D. An Equivalence Relation

gatecse-2006 set-theory&algebra normal relations

Answer key

4.12.28 Relations: GATE CSE 2007 | Question: 2



Let S be a set of n elements. The number of ordered pairs in the largest and the smallest equivalence relations on S are:

- A. n and n B. n^2 and n C. n^2 and 0 D. n and 1

gatecse-2007 set-theory&algebra normal relations

Answer key

4.12.29 Relations: GATE CSE 2009 | Question: 4



Consider the binary relation $R = \{(x, y), (x, z), (z, x), (z, y)\}$ on the set $\{x, y, z\}$. Which one of the following is **TRUE**?

- A. R is symmetric but NOT antisymmetric
 C. R is both symmetric and antisymmetric
 B. R is NOT symmetric but antisymmetric
 D. R is neither symmetric nor antisymmetric

gatecse-2009 set-theory&algebra easy relations

Answer key

4.12.30 Relations: GATE CSE 2010 | Question: 3



What is the possible number of reflexive relations on a set of 5 elements?

- A. 2^{10} B. 2^{15} C. 2^{20} D. 2^{25}

gatecse-2010 set-theory&algebra easy relations

Answer key

4.12.31 Relations: GATE CSE 2015 Set 2 | Question: 16



Let R be the relation on the set of positive integers such that aRb and only if a and b are distinct and let have a common divisor other than 1. Which one of the following statements about R is true?

- A. R is symmetric and reflexive but not transitive
 B. R is reflexive but not symmetric not transitive
 C. R is transitive but not reflexive and not symmetric
 D. R is symmetric but not reflexive and not transitive

gatecse-2015-set2 set-theory&algebra relations normal

Answer key

4.12.32 Relations: GATE CSE 2015 Set 3 | Question: 41



Let R be a relation on the set of ordered pairs of positive integers such that $((p, q), (r, s)) \in R$ if and only if $p - s = q - r$. Which one of the following is true about R ?

- A. Both reflexive and symmetric
 C. Not reflexive but symmetric
 B. Reflexive but not symmetric
 D. Neither reflexive nor symmetric

gatecse-2015-set3 set-theory&algebra relations normal

Answer key

4.12.33 Relations: GATE CSE 2016 Set 2 | Question: 26



A binary relation R on $\mathbb{N} \times \mathbb{N}$ is defined as follows: $(a, b)R(c, d)$ if $a \leq c$ or $b \leq d$. Consider the following propositions:

- $P : R$ is reflexive.
- $Q : R$ is transitive.

Which one of the following statements is **TRUE**?

- A. Both P and Q are true.
 C. P is false and Q is true.
 B. P is true and Q is false.
 D. Both P and Q are false.

gatecse-2016-set2 set-theory&algebra relations normal

Answer key

4.12.34 Relations: GATE CSE 2020 | Question: 17



Let $\mathcal{R}$ be the set of all binary relations on the set $\{1, 2, 3\}$. Suppose a relation is chosen from $\mathcal{R}$ at random. The probability that the chosen relation is reflexive (round off to 3 decimal places) is _____.

gatecse-2020 numerical-answers probability relations one-mark

Answer key

4.12.35 Relations: GATE CSE 2021 Set 1 | Question: 43



A relation R is said to be circular if aRb and bRc together imply cRa .

Which of the following options is/are correct?

- A. If a relation S is reflexive and symmetric, then S is an equivalence relation.
- B. If a relation S is circular and symmetric, then S is an equivalence relation.
- C. If a relation S is reflexive and circular, then S is an equivalence relation.
- D. If a relation S is transitive and circular, then S is an equivalence relation.

gatecse-2021-set1 multiple-selects set-theory&algebra relations two-marks

Answer key

4.12.36 Relations: GATE CSE 2025 | Set 2 | Question: 32



Let $\mathcal{F}$ be the set of all functions from $\{1, \dots, n\}$ to $\{0, 1\}$. Define the binary relation $\preceq$ on $\mathcal{F}$ as follows:

$\forall f, g \in \mathcal{F}, f \preceq g$ if and only if $\forall x \in \{1, \dots, n\}, f(x) \leq g(x)$, where $0 \leq 1$.

Which of the following statement(s) is/are TRUE?

- A. $\preceq$ is a symmetric relation
- B. $(\mathcal{F}, \preceq)$ is a partial order
- C. $(\mathcal{F}, \preceq)$ is a lattice
- D. $\preceq$ is an equivalence relation

gatecse2025-set2 set-theory&algebra functions relations multiple-selects two-marks

Answer key

4.12.37 Relations: GATE IT 2004 | Question: 4



Let R_1 be a relation from $A = \{1, 3, 5, 7\}$ to $B = \{2, 4, 6, 8\}$ and R_2 be another relation from B to $C = \{1, 2, 3, 4\}$ as defined below:

- i. An element x in A is related to an element y in B (under R_1) if $x + y$ is divisible by 3.
- ii. An element x in B is related to an element y in C (under R_2) if $x + y$ is even but not divisible by 3.

Which is the composite relation $R_1 R_2$ from A to C ?

- A. $R_1 R_2 = \{(1, 2), (1, 4), (3, 3), (5, 4), (7, 3)\}$
- B. $R_1 R_2 = \{(1, 2), (1, 3), (3, 2), (5, 2), (7, 3)\}$
- C. $R_1 R_2 = \{(1, 2), (3, 2), (3, 4), (5, 4), (7, 2)\}$
- D. $R_1 R_2 = \{(3, 2), (3, 4), (5, 1), (5, 3), (7, 1)\}$

gateit-2004 set-theory&algebra relations normal

Answer key

4.13

Set Theory (27)

4.13.1 Set Theory: GATE CSE 1993 | Question: 17



Out of a group of 21 persons, 9 eat vegetables, 10 eat fish and 7 eat eggs. 5 persons eat all three. How many persons eat at least two out of the three dishes?

gate1993 set-theory&algebra easy set-theory descriptive

Answer key

4.13.2 Set Theory: GATE CSE 1993 | Question: 8.3



Let S be an infinite set and $S_1, \dots, S_n$ be sets such that $S_1 \cup S_2 \cup \dots \cup S_n = S$. Then

- A. at least one of the sets S_i is a finite set
- B. not more than one of the sets S_i can be finite
- C. at least one of the sets S_i is an infinite
- D. not more than one of the sets S_i can be infinite
- E. None of the above

gate1993 set-theory&algebra normal set-theory

Answer key

4.13.3 Set Theory: GATE CSE 1993 | Question: 8.4



Let A be a finite set of size n . The number of elements in the power set of $A \times A$ is:

- A. 2^{2^n}
- B. 2^{n^2}
- C. $(2^n)^2$
- D. $(2^2)^n$
- E. None of the above

gate1993 set-theory&algebra easy set-theory

Answer key

4.13.4 Set Theory: GATE CSE 1994 | Question: 2.4



The number of subsets $\{1, 2, \dots, n\}$ with odd cardinality is _____

gate1994 set-theory&algebra easy set-theory fill-in-the-blanks

Answer key

4.13.5 Set Theory: GATE CSE 1995 | Question: 1.20



The number of elements in the power set $P(S)$ of the set $S = \{\{\emptyset\}, 1, \{2, 3\}\}$ is:

- A. 2
- B. 4
- C. 8
- D. None of the above

gate1995 set-theory&algebra normal set-theory

Answer key

4.13.6 Set Theory: GATE CSE 1995 | Question: 25b



Determine the number of positive integers (≤ 720) which are not divisible by any of 2, 3 or 5.

gate1995 set-theory&algebra set-theory numerical-answers

Answer key

4.13.7 Set Theory: GATE CSE 1996 | Question: 1.1



Let A and B be sets and let A^c and B^c denote the complements of the sets A and B . The set $(A - B) \cup (B - A) \cup (A \cap B)$ is equal to

- A. $A \cup B$
- B. $A^c \cup B^c$
- C. $A \cap B$
- D. $A^c \cap B^c$

gate1996 set-theory&algebra easy set-theory

Answer key

4.13.8 Set Theory: GATE CSE 1998 | Question: 2.4



In a room containing 28 people, there are 18 people who speak English, 15 people who speak Hindi and 22 people who speak Kannada. 9 persons speak both English and Hindi, 11 persons speak both Hindi and Kannada whereas 13 persons speak both Kannada and English. How many speak all three languages?

- A. 9
- B. 8
- C. 7
- D. 6

gate1998 set-theory&algebra easy set-theory

Answer key

4.13.9 Set Theory: GATE CSE 2000 | Question: 2.6



Let $P(S)$ denotes the power set of set S . Which of the following is always true?

- A. $P(P(S)) = P(S)$
- B. $P(S) \cap P(P(S)) = \{\emptyset\}$
- C. $P(S) \cap S = P(S)$
- D. $S \notin P(S)$

gatecse-2000 set-theory&algebra easy set-theory

Answer key

4.13.10 Set Theory: GATE CSE 2000 | Question: 6



Let S be a set of n elements $\{1, 2, \dots, n\}$ and G a graph with 2^n vertices, each vertex corresponding to a distinct subset of S . Two vertices are adjacent iff the symmetric difference of the corresponding sets has exactly 2 elements. Note: The symmetric difference of two sets R_1 and R_2 is defined as $(R_1 \setminus R_2) \cup (R_2 \setminus R_1)$

Every vertex in G has the same degree. What is the degree of a vertex in G ?

How many connected components does G have?

gatecse-2000 set-theory&algebra normal descriptive set-theory

Answer key

4.13.11 Set Theory: GATE CSE 2001 | Question: 2.2



Consider the following statements:

- S_1 : There exists infinite sets A, B, C such that $A \cap (B \cup C)$ is finite.
- S_2 : There exists two irrational numbers x and y such that $(x + y)$ is rational.

Which of the following is true about S_1 and S_2 ?

- A. Only S_1 is correct
- B. Only S_2 is correct
- C. Both S_1 and S_2 are correct
- D. None of S_1 and S_2 is correct

gatecse-2001 set-theory&algebra normal set-theory

Answer key

4.13.12 Set Theory: GATE CSE 2001 | Question: 3



- a. Prove that $\text{powerset}(A \cap B) = \text{powerset}(A) \cap \text{powerset}(B)$
- b. Let $\text{sum}(n) = 0 + 1 + 2 + \dots + n$ for all natural numbers n . Give an induction proof to show that the following equation is true for all natural numbers m and n :

$$\text{sum}(m + n) = \text{sum}(m) + \text{sum}(n) + mn$$

gatecse-2001 set-theory&algebra normal set-theory descriptive

Answer key

4.13.13 Set Theory: GATE CSE 2005 | Question: 8



Let A, B and C be non-empty sets and let $X = (A - B) - C$ and $Y = (A - C) - (B - C)$. Which one of the following is TRUE?

- A. $X = Y$
- B. $X \subset Y$
- C. $Y \subset X$
- D. None of these

gatecse-2005 set-theory&algebra easy set-theory

Answer key

4.13.14 Set Theory: GATE CSE 2006 | Question: 22



Let E, F and G be finite sets. Let

- $X = (E \cap F) - (F \cap G)$ and
- $Y = (E - (E \cap G)) - (E - F)$.

Which one of the following is true?

- A. $X \subset Y$
C. $X = Y$

- B. $X \supset Y$
D. $X - Y \neq \emptyset$ and $Y - X \neq \emptyset$

gatecse-2006 set-theory&algebra normal set-theory

Answer key

4.13.15 Set Theory: GATE CSE 2006 | Question: 24



Given a set of elements $N = 1, 2, \dots, n$ and two arbitrary subsets $A \subseteq N$ and $B \subseteq N$, how many of the $n!$ permutations π from N to N satisfy $\min(\pi(A)) = \min(\pi(B))$, where $\min(S)$ is the smallest integer in the set of integers S , and $\pi(S)$ is the set of integers obtained by applying permutation π to each element of S ?

- A. $(n - |A \cup B|)|A||B|$
C. $n! \frac{|A \cap B|}{|A \cup B|}$
B. $(|A|^2 + |B|^2)n^2$
D. $\frac{|A \cap B|^2}{{}^nC_{|A \cup B|}}$

gatecse-2006 set-theory&algebra normal set-theory

Answer key

4.13.16 Set Theory: GATE CSE 2008 | Question: 2



If P, Q, R are subsets of the universal set U , then

$$(P \cap Q \cap R) \cup (P^c \cap Q \cap R) \cup Q^c \cup R^c$$

is

- A. $Q^c \cup R^c$
C. $P^c \cup Q^c \cup R^c$
B. $P \cup Q^c \cup R^c$
D. U

gatecse-2008 normal set-theory&algebra set-theory

Answer key

4.13.17 Set Theory: GATE CSE 2014 Set 2 | Question: 50



Consider the following relation on subsets of the set S of integers between 1 and 2014. For two distinct subsets U and V of S we say $U < V$ if the minimum element in the symmetric difference of the two sets is in U .

Consider the following two statements:

- $S1$: There is a subset of S that is larger than every other subset.
- $S2$: There is a subset of S that is smaller than every other subset.

Which one of the following is CORRECT?

- A. Both $S1$ and $S2$ are true
C. $S2$ is true and $S1$ is false
B. $S1$ is true and $S2$ is false
D. Neither $S1$ nor $S2$ is true

gatecse-2014-set2 set-theory&algebra normal set-theory

Answer key

4.13.18 Set Theory: GATE CSE 2015 Set 1 | Question: 16



For a set A , the power set of A is denoted by 2^A . If $A = \{5, \{6\}, \{7\}\}$, which of the following options are TRUE?

- I. $\emptyset \in 2^A$
II. $\emptyset \subseteq 2^A$
III. $\{5, \{6\}\} \in 2^A$
IV. $\{5, \{6\}\} \subseteq 2^A$

- A. I and III only
B. II and III only
C. I, II and III only
D. I, II and IV only

gatecse-2015-set1 set-theory&algebra set-theory normal

Answer key

4.13.19 Set Theory: GATE CSE 2015 Set 2 | Question: 18



The cardinality of the power set of $\{0, 1, 2, \dots, 10\}$ is _____

gatecse-2015-set2 set-theory&algebra set-theory easy numerical-answers

Answer key

4.13.20 Set Theory: GATE CSE 2015 Set 3 | Question: 23



Suppose U is the power set of the set $S = \{1, 2, 3, 4, 5, 6\}$. For any $T \in U$, let $|T|$ denote the number of elements in T and T' denote the complement of T . For any $T, R \in U$ let $T \setminus R$ be the set of all elements in T which are not in R . Which one of the following is true?

- A. $\forall X \in U, (|X| = |X'|)$
- B. $\exists X \in U, \exists Y \in U, (|X| = 5, |Y| = 5 \text{ and } X \cap Y = \phi)$
- C. $\forall X \in U, \forall Y \in U, (|X| = 2, |Y| = 3 \text{ and } X \setminus Y = \phi)$
- D. $\forall X \in U, \forall Y \in U, (X \setminus Y = Y' \setminus X')$

gatecse-2015-set3 set-theory&algebra set-theory normal

Answer key

4.13.21 Set Theory: GATE CSE 2016 Set 2 | Question: 28



Consider a set U of 23 different compounds in a chemistry lab. There is a subset S of U of 9 compounds, each of which reacts with exactly 3 compounds of U . Consider the following statements:

- I. Each compound in $U \setminus S$ reacts with an odd number of compounds.
- II. At least one compound in $U \setminus S$ reacts with an odd number of compounds.
- III. Each compound in $U \setminus S$ reacts with an even number of compounds.

Which one of the above statements is **ALWAYS TRUE**?

- A. Only I
- B. Only II
- C. Only III
- D. None.

gatecse-2016-set2 set-theory&algebra difficult set-theory

Answer key

4.13.22 Set Theory: GATE CSE 2017 Set 1 | Question: 47



The number of integers between 1 and 500 (both inclusive) that are divisible by 3 or 5 or 7 is _____.

gatecse-2017-set1 set-theory&algebra normal numerical-answers set-theory

Answer key

4.13.23 Set Theory: GATE CSE 2021 Set 2 | Question: 37



For two n -dimensional real vectors P and Q , the operation $s(P, Q)$ is defined as follows:

$$s(P, Q) = \sum_{i=1}^n (P[i] \cdot Q[i])$$

Let $\mathcal{L}$ be a set of 10-dimensional non-zero real vectors such that for every pair of distinct vectors $P, Q \in \mathcal{L}$, $s(P, Q) = 0$. What is the maximum cardinality possible for the set $\mathcal{L}$?

- A. 9
- B. 10
- C. 11
- D. 100

gatecse-2021-set2 set-theory&algebra set-theory two-marks

Answer key

4.13.24 Set Theory: GATE IT 2004 | Question: 2



In a class of 200 students, 125 students have taken Programming Language course, 85 students have taken Data Structures course, 65 students have taken Computer Organization course; 50 students have taken both Programming Language and Data Structures, 35 students have taken both Programming Language and Computer Organization; 30 students have taken both Data Structures and Computer Organization, 15 students have taken all the three courses.

How many students have not taken any of the three courses?

A. 15

B. 20

C. 25

D. 30

gateit-2004 set-theory&algebra easy set-theory

Answer key

4.13.25 Set Theory: GATE IT 2005 | Question: 33



Let A be a set with n elements. Let C be a collection of distinct subsets of A such that for any two subsets S_1 and S_2 in C , either $S_1 \subset S_2$ or $S_2 \subset S_1$. What is the maximum cardinality of C ?

A. n

B. $n + 1$

C. $2^{n-1} + 1$

D. $n!$

gateit-2005 set-theory&algebra normal set-theory

Answer key

4.13.26 Set Theory: GATE IT 2006 | Question: 23



Let P , Q and R be sets let Δ denote the symmetric difference operator defined as $P\Delta Q = (P \cup Q) - (P \cap Q)$. Using Venn diagrams, determine which of the following is/are TRUE?

- I. $P\Delta(Q \cap R) = (P\Delta Q) \cap (P\Delta R)$
- II. $P \cap (Q \cap R) = (P \cap Q)\Delta(P\Delta R)$

A. I only

B. II only

C. Neither I nor II

D. Both I and II

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Answer key

4.13.27 Set Theory: GATE IT 2006 | Question: 24



What is the cardinality of the set of integers X defined below?

$X = \{n \mid 1 \leq n \leq 123, n \text{ is not divisible by either } 2, 3 \text{ or } 5\}$

A. 28

B. 33

C. 37

D. 44

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Answer key

Answer Keys

4.1.1	N/A	4.1.2	N/A	4.1.3	C	4.1.4	D	4.1.5	A
4.1.6	A	4.1.7	B	4.1.8	A	4.2.1	True	4.2.2	D
4.3.1	N/A	4.3.2	N/A	4.3.3	N/A	4.3.4	B	4.3.5	A
4.3.6	C	4.3.7	N/A	4.3.8	C	4.3.9	A	4.3.10	N/A
4.3.11	A	4.3.12	B	4.3.13	B	4.3.14	D	4.3.15	B
4.3.16	C	4.3.17	C	4.3.18	16	4.3.19	D	4.3.20	B
4.3.21	A	4.3.22	36	4.3.23	0.95	4.3.24	2	4.3.25	B;C
4.3.26	B;C;D	4.3.27	2	4.3.28	C	4.3.29	D	4.4.1	N/A
4.4.2	N/A	4.4.3	N/A	4.4.4	N/A	4.4.5	C	4.4.6	D
4.4.7	N/A	4.4.8	C	4.4.9	C	4.4.10	D	4.4.11	N/A
4.4.12	N/A	4.4.13	N/A	4.4.14	B	4.4.15	A	4.4.16	D
4.4.17	C	4.4.18	A	4.4.19	C	4.4.20	A	4.4.21	A
4.4.22	C	4.4.23	A	4.4.24	5	4.4.25	4	4.4.26	42
4.4.27	B	4.4.28	7	4.4.29	B	4.4.30	A;B;C	4.4.31	A;D
4.4.32	B;D	4.4.33	4	4.5.1	B	4.6.1	N/A	4.6.2	N/A

4.6.3	D
4.6.8	D
4.8.1	24
4.8.6	C
4.10.3	B
4.10.8	C
4.11.3	A
4.12.4	N/A
4.12.9	N/A
4.12.14	N/A
4.12.19	N/A
4.12.24	B
4.12.29	D
4.12.34	0.1249:0.1259
4.13.2	C
4.13.7	A
4.13.12	N/A
4.13.17	A
4.13.22	271
4.13.27	B

4.6.4	5
4.6.9	0
4.8.2	8
4.8.7	N/A
4.10.4	A
4.10.9	5
4.11.4	8
4.12.5	N/A
4.12.10	N/A
4.12.15	N/A
4.12.20	C
4.12.25	C
4.12.30	C
4.12.35	C
4.13.3	B
4.13.8	D
4.13.13	A
4.13.18	C
4.13.23	B

4.6.5	B
4.6.10	A
4.8.3	36
4.9.1	D
4.10.5	N/A
4.10.10	D
4.12.1	False
4.12.6	N/A
4.12.11	A
4.12.16	B
4.12.21	B
4.12.26	D
4.12.31	D
4.12.36	B;C
4.13.4	N/A
4.13.9	B
4.13.14	C
4.13.19	2048
4.13.24	C

4.6.6	N/A
4.7.1	N/A
4.8.4	D
4.10.1	N/A
4.10.6	D
4.11.1	B
4.12.2	N/A
4.12.7	D
4.12.12	B
4.12.17	C
4.12.22	D
4.12.27	A
4.12.32	C
4.12.37	C
4.13.5	C
4.13.10	N/A
4.13.15	C
4.13.20	D
4.13.25	B

4.6.7	B
4.7.2	N/A
4.8.5	D
4.10.2	E
4.10.7	A
4.11.2	D
4.12.3	N/A
4.12.8	B
4.12.13	C
4.12.18	A;C
4.12.23	N/A
4.12.28	B
4.12.33	B
4.13.1	N/A
4.13.6	192
4.13.11	C
4.13.16	D
4.13.21	B
4.13.26	C